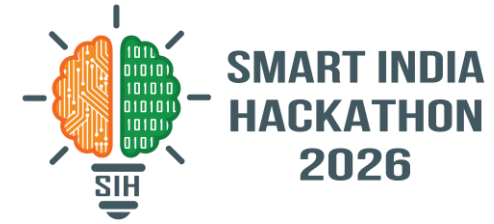


SMART INDIA HACKATHON 2026



- **Problem Statement ID** - SIH26094
- **Problem Statement Title** - AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities
- **Theme** - MedTech / BioTech / HealthTech
- **PS Category** - Software
- **Ministry** - Ministry of Social Justice and Empowerment (MoSJE)
- **Team ID** - 141444
- **Team Name (Registered on portal)** - DEKO
- **College** - University College of Engineering, Panruti
- **Team Leader** - BALAJI P

Description:

- Victims of atrocities frequently experience prolonged psychological distress after complaint registration due to threats, intimidation, repeated court appearances, delays in investigation/trial, social ostracism, economic hardship, and rehabilitation challenges.
- Existing mechanisms focus primarily on post-hoc legal and financial compensation, leaving a critical void in continuous monitoring of victim well-being.
- Proposed Technological Solution: An AI-powered Dynamic Mental Health Monitoring & Distress Prediction System (KEFI AI) integrated with NHAA (14566), IVRS calls, SMS, mobile applications, and web portal.
- Key Mandate: Continuously monitor psychological well-being, analyze multimodal signals (voice prosody, NLP sentiment, behavior), compute a Dynamic Distress Score (0–100), forecast crises 72 hours before escalation, and automatically alert District Magistrates (DM), SPs, and counsellors.

- **KEFI AI Identity & Greek Origin:** KEFI AI (Kinematic Emotion & Forecasting Intervention System) — derived from Kéfi (Κέφι): spirit of joy, emotional vitality, passion & inner healing after trauma.
- **Continuous Post-Complaint Monitoring:** Proactively monitors crime victims (SC/ST PoA Act 1989 / NHAA 14566) across investigation, court trial, compensation, and rehabilitation phases rather than waiting for crisis requests.
- **Multichannel Interaction Hub:** Periodic check-ins via IVRS 14566 calls, Chatbot, SMS, Mobile App, and Web Portal with digital OTP consent and end-to-end AES-256 encryption.
- **Dual-Risk Assessment Engine:** 1) Immediate Physical Threat Check: Screens for active intimidation language. 2) Dynamic Distress Score (DDS 0–100): Multi-signal mathematical metric ($DDS = w1 * E + w2 * V + w3 * B + w4 * H + w5 * C + w6 * T$).
- **72-Hour Longitudinal Crisis Forecasting:** Calculates distress velocity ($dDDS/dt$) over past 7 days to predict psychological crisis 72 hours before escalation occurs.
- **3-Tier Risk Bands & Action Matrix:** ● Green/Yellow (0–30): Routine wellness check-ins & self-help tools. ● Orange (31–60): Focused check-ins & scheduled tele-counselling. ● Red (61–100): Real-time alert to District Magistrate (DM) / SP dashboard, witness protection & crisis team dispatch.

Multichannel Ingestion (14566)

IVRS 14566, Chatbot, SMS, Web & Mobile App with OTP verification & DPDP consent.

Emotion Wheel & Voice AI

Granular Plutchik emotion wheel for non-verbal input + vocal jitter & pitch tremor analysis.

Calm Space & Silent Mode

Box Breathing (4-4-4-4), 5-4-3-2-1 Grounding, Ambient Audio + Silent Mode posture tracking.

4 Role-Based Logins

1. Victim / Witness Portal
2. Field Officer App
3. DM / SP Dashboard
4. National MoSJE Admin

Idea / Approach Flowchart

Step 1: Onboarding & Multichannel Ingestion: Victim complaint registered -> OTP verification -> Digital consent (DPDP Act 2023) -> Scheduled outreach via IVRS 14566, Chatbot, SMS, or Mobile App.

Step 2: Multimodal AI Signal Ingestion: NLP Text Sentiment (Transformer BERT for fear valence & threat keywords 'dhamki', 'withdraw'), Vocal Acoustic Prosody (pitch tremor Hz, jitter %, pause duration), and Behavioral Check-in Gaps.

Step 3: Dynamic Distress Scoring (0–100): Calculates continuous formula: $DDS = 0.25 * E + 0.20 * V + 0.15 * B + 0.15 * H + 0.10 * C + 0.15 * T$.

Step 4: 72h Crisis Forecast & Explainable AI (XAI): Time-series trend velocity (dDDS/dt) predicts crisis 72h prior. SHAP/LIME point attributions (+21 threat, +18 fear, +14 velocity, +12 missed check-ins, +9 voice stress).

Step 5: Automated 3-Tier Risk Action Dispatch: ● 0–30: Self-help grounding audio & rights pamphlets | ● 31–60: Focused check-ins & tele-counselling | ● 61–100: Real-time alert to DM / SP dashboard, police protection & crisis team dispatch.

Parameter	Existing Manual System	Proposed KEFI AI Platform
Monitoring Model	Passive (Victim must ask for help during severe crisis)	Proactive & Continuous (Automated scheduled check-ins)
Distress Assessment	Subjective / Nil post-complaint registration	Mathematical Dynamic Distress Score (DDS: 0–100)
Data Modalities	Manual paper records / infrequent police visits	Multimodal (Text NLP, Acoustic Prosody, Behavior, Threat Keywords)
Early Warning	Reactive (Intervention only after suicide / harm)	72-Hour Predictive Trend Velocity Forecasting (dDDS/dt)
Decision Transparency	Black-box officer discretion	Explainable AI (SHAP/LIME) with exact numerical point additions
Governance Visibility	Disconnected district registers	Real-Time District (DM/SP) & National MoSJE Dashboards

Analysis of the Feasibility:

- **Seamless Operational Integration:** Integrates directly with existing MoSJE National Helpline Against Atrocities (NHAA 14566), District Magistrates (DM), SPs, and welfare officers without adding administrative workload.
- **Hardware-Agnostic & Scalable:** Requires no costly wearable hardware. Operates via standard phone calls (IVRS), SMS, and web portal.
- **Low-Bandwidth Adaptability:** Low Internet Mode allows offline grounding audio, low-data layout, and basic SMS fallback for remote rural atrocity victims.

Potential Challenges & Strategies for Overcoming:

- **Risk 1: Alarm Fatigue & False Positives**
↳ Strategy: Dual-layer verification combining automated AI Distress Score with Field Welfare Officer ground observation notes.
- **Risk 2: Data Privacy & Victim Security**
↳ Strategy: 100% DPDP Act 2023 compliant with end-to-end AES-256 encryption, OTP consent, and victim data purge controls.
- **Risk 3: Literacy & Multilingual Barriers**
↳ Strategy: Multilingual conversational AI supporting Tanglish (தமிழ்-English), Tamil, Hindi, and English
+ Silent Mode non-verbal support.

TECHNOLOGICAL STACK:

- **Frontend Architecture:** React 18, Vite 8, Vanilla CSS & Tailwind, Web Speech API (Voice Recognition & TTS), face-api.js for Silent Vision Detection.
- **Backend Microservices:** Python 3.12, FastAPI, Uvicorn ASGI Server, RESTful API architecture with <250ms response latency.
- **AI / NLP & Acoustic Models:** Hugging Face Transformers (BERT Sentiment for fear valence), Acoustic Speech Prosody Analyzer (Pitch Tremor Hz, Jitter %, Pause Gaps), Cognitive Distortion Mapper.
- **Predictive & Explainable AI (XAI):** Time-Series Distress Velocity Forecasting ($dDDS/dt$ for 72h crisis prediction), SHAP & LIME Hybrid Point Attribution Engine (+21, +18, +14, +12, +9).
- **Database & Cloud Infrastructure:** SQLite / PostgreSQL with SQLAlchemy ORM, Redis for session caching, Docker containerization for MeghRaj / NIC Cloud deployment.
- **Security & Governance:** Digital Personal Data Protection (DPDP) Act 2023 Compliance, AES-256 Data Encryption at rest & in transit, SHA-256 OTP Authentication, Role-Based Access Control (RBAC).

Victim Portal & Emotion Wheel

Granular emotion wheel, daily mood log & instant calming support.

Dynamic Distress (0-100) & Trend

Continuous score tracking with 72h crisis velocity forecasting.

DM / SP Alert & Intervention

Real-time red-band crisis alerts, witness protection & officer dispatch.

Calm Space & Silent Sanctuary

Box Breathing (4-4-4-4), 5-4-3-2-1 Grounding & ambient audio.

Potential Impact on the Target Audience:

- **Victims of Rape, Gang Rape & Grievous Hurt:** Continuous psychological care reduces trauma isolation and prevents severe PTSD, self-harm, and psychological collapse.
- **Witnesses Facing Intimidation & Threats:** Proactive screening of coercion keywords and immediate witness protection prevent trial retraction and intimidation.
- **Families Affected by Caste-Based Violence:** Systematic monitoring of relief distribution, legal aid progress, and rehabilitation milestones under SC/ST PoA Act.
- **Victim Empowerment & Autonomy:** Equipped with self-help grounding tools, box breathing guides, legal rights pamphlets, and transparent feedback channels.

Benefits of the Solution (Social, Legal, Administrative):

- **Early Crisis Detection & Prevention:** Intercepts acute distress 72 hours before crisis, enabling timely tele-counselling and suicide prevention.
- **Strengthened Justice System Trust:** Proactive state care rebuilds victim faith, drastically reducing trial dropouts and witness hostility in court.
- **Multi-Agency Synergy:** Unifies welfare departments, police liaisons, and tele-counsellors under a shared real-time incident workflow.
- **Evidence-Based Governance:** State and National MoSJE heatmaps provide administrators and District Magistrates data-driven insights on atrocity hotspots.
- **Reduced Administrative Burden:** Automated multichannel check-ins and AI triage cut manual welfare follow-up overhead by over 70%.

Reference Links & Research Work:

- **Statutory Act:** Scheduled Castes and the Scheduled Tribes (Prevention of Atrocities) Act, 1989 & PoA Rules 1995: <https://socialjustice.gov.in>
- **National Helpline:** MoSJE National Helpline Against Atrocities (NHAA 14566) Integration: <https://socialjustice.gov.in/schemes/nhaa>
- **Data Privacy Law:** Digital Personal Data Protection (DPDP) Act, 2023 — Government of India: <https://www.meity.gov.in/dpdp-act>
- **Clinical Research:** Scherer, K. R., et al. (2013). Acoustic Markers of Speech Under Psychological Stress. Computer Speech & Language.
- **Explainable AI:** Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions (SHAP). NeurIPS.
- **Project Repository:** KEFI AI Full Source Code & Web Application: <https://github.com/balajikrishnan031/SIH-KEFI>

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